Uncompahgre PIT Tag Array Data Visualization How-To

Sam Graf

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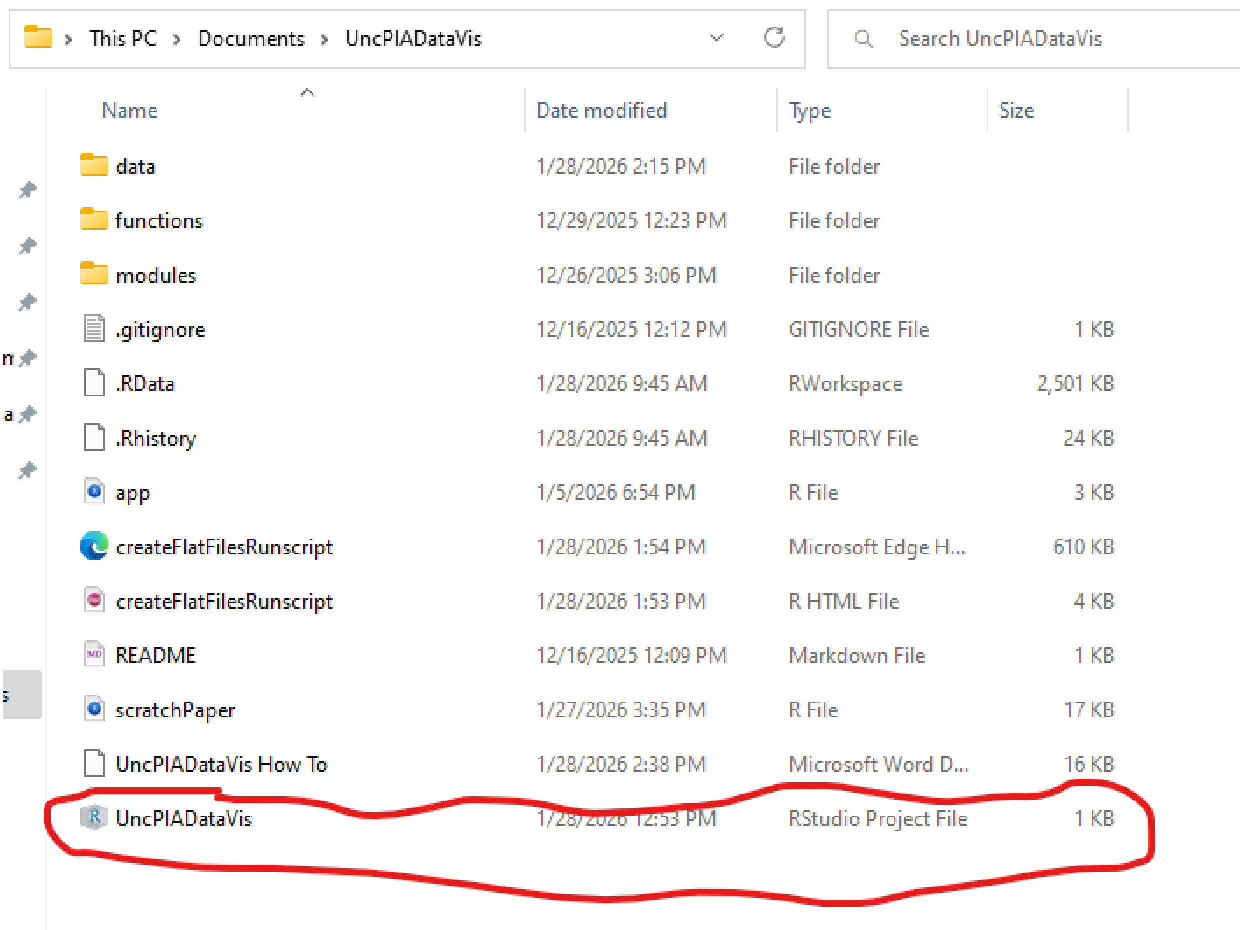
Location: Lives with Dan

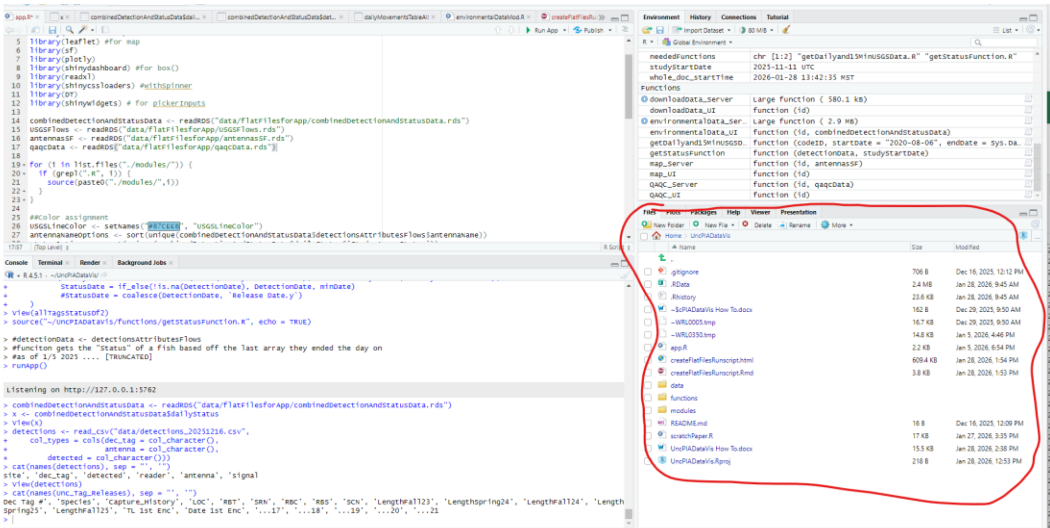
This document is intended as guidance to visualize data from the Uncompahgre PIT array and update with new data as needed.

To Open/Run

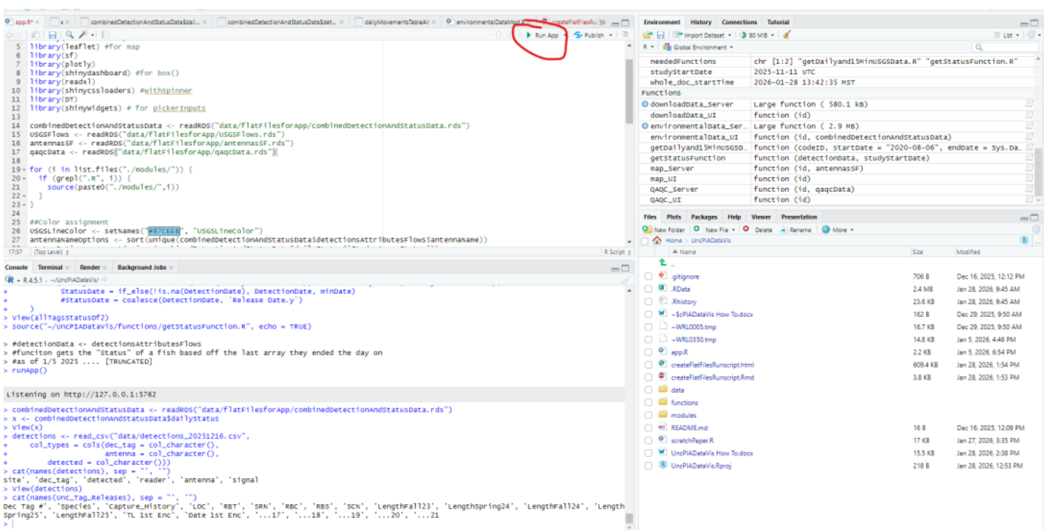
- Navigate to the folder where the app is located and open the .rproj file. This will automatically open RStudio and set your project directory relative to the .rpoj file which is important for reading in files.



- Once Rstudio is open, use the file pane on the lower right to navigate and open app.r (the app itself). If you need to update data, open “createFlatFilesRunscript.rmd” (used to update data in the app).



- If app.R is open and you don’t need to update data, run the app using the “run app” button in the top right. You will first probably be prompted to download/install required packages if this is the first time running the app.



Updating App Workflow

1. Collect/update manual data inputs
2. Run R markdown script
    a. This combines the data inputs and extrapolates fish “status” for missing rows
    b. Creates static data files to reduce calculations and improve “smoothness” within the app
3. Run app

Data Inputs

Manually Updated

The script currently takes 3 tabular data files from external sources. They are located in the "data" folder of the project directory. When updating the script, these data will have to be manually updated and placed in the "data" directory.

- antennaMetadata.xlsx
  - contains antenna numbers/names and lat/long cords
  - used to turn antenna numbers coming off detections file into Upstream/downstream arrays, also to make sf object of arrays for map
  - needed columns: antennaNumber, antennaName, lat, long
- detections_relevantDate.csv
  - detections file from Biomark website with tags in Dec format
  - Needed column names (order doesn't matter): 'dec_tag', 'detected', 'antenna',
- Unc_Tag_Releases_relevantDate.csv
  - All tags with species, length, and release data
  - Prepared by Dan
  - Needed column names (order doesn't matter): 'Dec Tag #', 'Species', 'TL 1st Enc', 'Date 1st Enc'

Automatically Updated

The script gathers daily data from USGS gages using the "DataRetrieval" package in R starting at the first date of antenna detections. Upon app re-run, the data will automatically update to the current date for available data. The gage used is:

- Uncompahgre River Below Ridgway Reservoir (09147025)

Running RMarkdown Script/Updating the App

- Put new manual tabular data inputs in the "data" folder of the project directory
  - Make sure that data is correctly formatted including file type and needed columns.
- Open "createFlatFilesRunscript.Rmd" from the files pane within Rstudio within the project directory (see "To Open/Run" if you haven't opened the .rproject file yet)
- If the new updated tabular data have different filenames, update the new name in the RMarkdown document, found in the first chunk towards the top of the document.

```r
```{r setup, include=FALSE}
whole_doc_startTime <- Sys.time()

library(tidyverse)
library(dataRetrieval)
library(readxl)
library(sf)

neededFunctions <- c("getDailyand15MinUSGSData.R", "getStatusFunction.R")

for (i in neededFunctions) {
  source(paste0("./functions/",i))
}

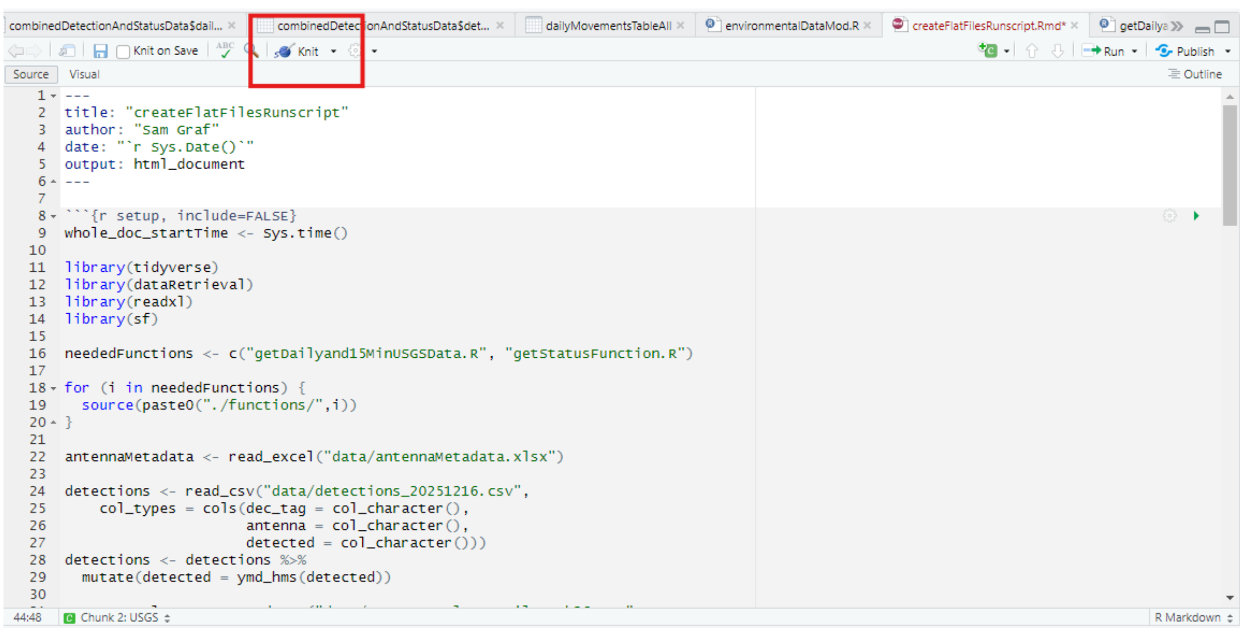
antennaMetadata <- read_excel("data/antennaMetadata.xlsx")

detections <- read_csv("data/detections_20251216.csv",
    col_types = cols(dec_tag = col_character(),
                     antenna = col_character(),
                     detected = col_character()))
detections <- detections %>%
  mutate(detected = ymd_hms(detected))

Unc_Tag_Releases <- read_csv("data/Unc Tag Release File Feb26.csv",
    col_types = cols(`Dec Tag #` = col_character()))

antennasSFAll <- st_as_sf(antennaMetadata, coords = c("long", "lat"), crs = 4326)
```

- Run the whole document by clicking "knit" in the top of the coding pane. If it's your first time running the script, follow the prompts to install the required packages.



- The document will produce an .html file if successful. Static files are updated and saved in the data directory and the app is ready to run (see "To Open/Run".
```

createFlatFilesRunscript

Sam Graf

2026-01-28

The whole document took 0.13 minutes to run. Flat Files are saved to the flat files directory and the app is ready to run with the most updated data.

QAQC

- There are some automatic QAQC checks in the runscript, including checking for duplicate tag entries in the release file, checking to make sure Dec Tag formatting is correct in the detection and release files (length of each tag as a string is 16, with 3 digits preceding the decimal and 12 digits proceeding), and making sure each tag detected has release data (Species, Length, Release date) associated with it. If one of these checks are completed and something is abnormal, the runscript will reflect this by displaying messages and related table. The app will probably still run with this incorrect data but it’s best to correct the data before making any interpretations etc. Below is an example of QAQC in the runscript, tag number is fabricated.

Tags in the Release Data are not the correct format. Length of all Dec Tag entries shoult be 16: 3 digits before decimal, 12 subsequent digits.

Show 10 entries Search:

String Lengths of Tags Release Data (should be 16)

String Length of Released Tag Entries	Count
16	2440
18	1
	1

Showing 1 to 3 of 3 entries Previous 1 Next

There are tags with duplicate entries in the release file.

Show 10 entries Search:

Tags in release file that have more than 1 entry in the release file.

Dec Tag #	Number of Tag Entries
989.001030620161	2

Showing 1 to 1 of 1 entries Previous 1 Next

There are detected tags without release data.

Show 10 entries Search:

Detected tags without release data.

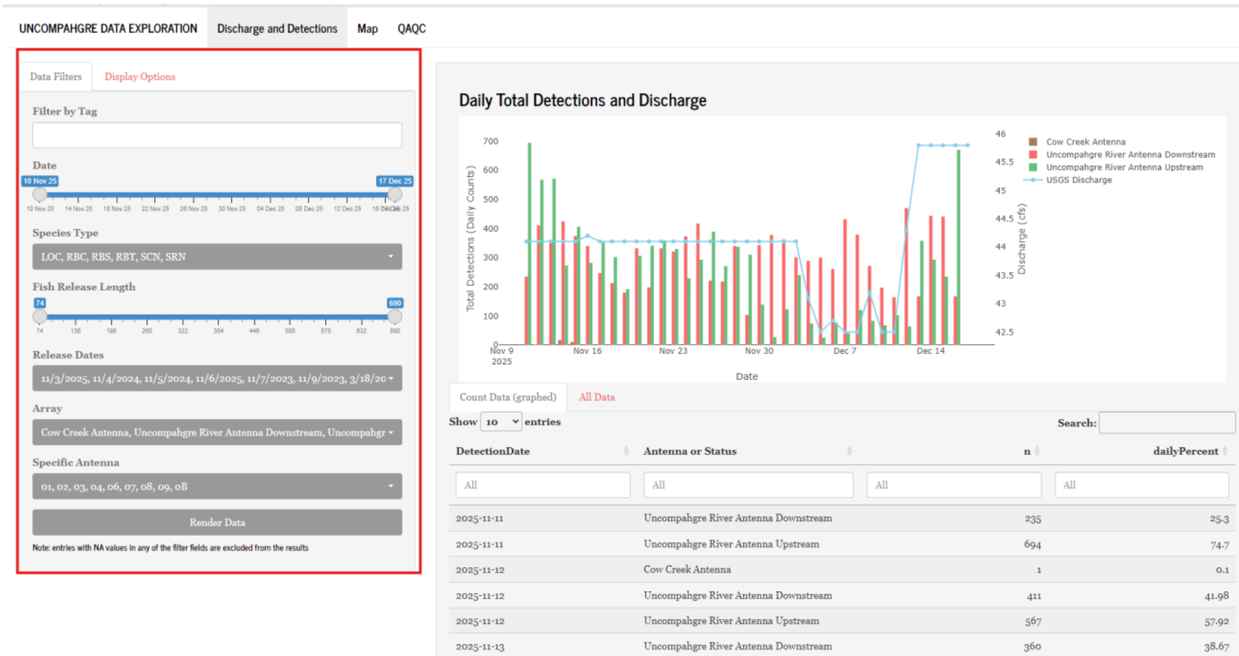
site	dec_tag	detected	reader	antenna	signal	antennaName	geometry	Release Date	Species	TL 1st Enc. (mm)	DetectionDate	Flow
	989.101030620161	2025-11-11T00:02:53Z		02		Uncompahgre River Antenna Upstream	[object Object]				2025-11-11	44.1

Navigating the App

Discharge and Detections

Data Filters

- Use the filters in the left sidebar to select preferred data, then click “Render data” to display it.



Display Options

- Click "Display Options" tab in the sidebar to change how the plot is displayed. There is the option of "Total detections" or "Status"

Status Definition

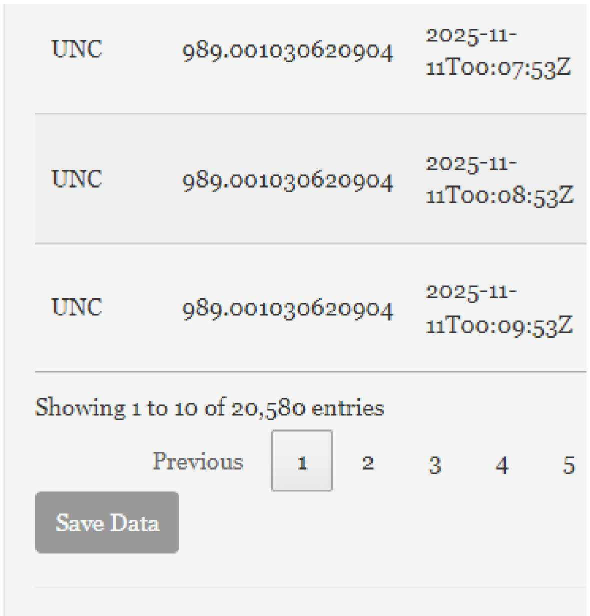
- Where the fish ended the day determines their "Status". If a fish ended with a detection on the upstream array, it is considered to be "within the study area". If a fish ended the day with a detection on the downstream array, it is considered to be "outside the study area". Cow creek detections are treated separately.
- A fish is assumed to remain in that status for ensuing days until a new detection is recorded.
- Fish without any detections are assumed to be within the study area
- If a fish was released but then the first detection of its history was on the downstream array, it is assumed that pre-study, the fish was assumed to be outside the study area and remained so for dates of the study until its first detection
  - Example: 989.001040500060

Plot

- The plot displayed is an interactive daily histogram of detections or fish status (see "Display options") overlaid with USGS flow data from the Ridgway gage. Hovering will show data associated with each trace on the graph.
- Double click a trace in the legend to isolate that trace, or click the trace just once to remove it from the display
- Save the graph as a .png with "Download plot as a png" option in the top right of the plot by the legend
- Zoom in by clicking and dragging or using the option in the top right, and zoom out using options in the top right including "autoscale" to get back to normal.

Table

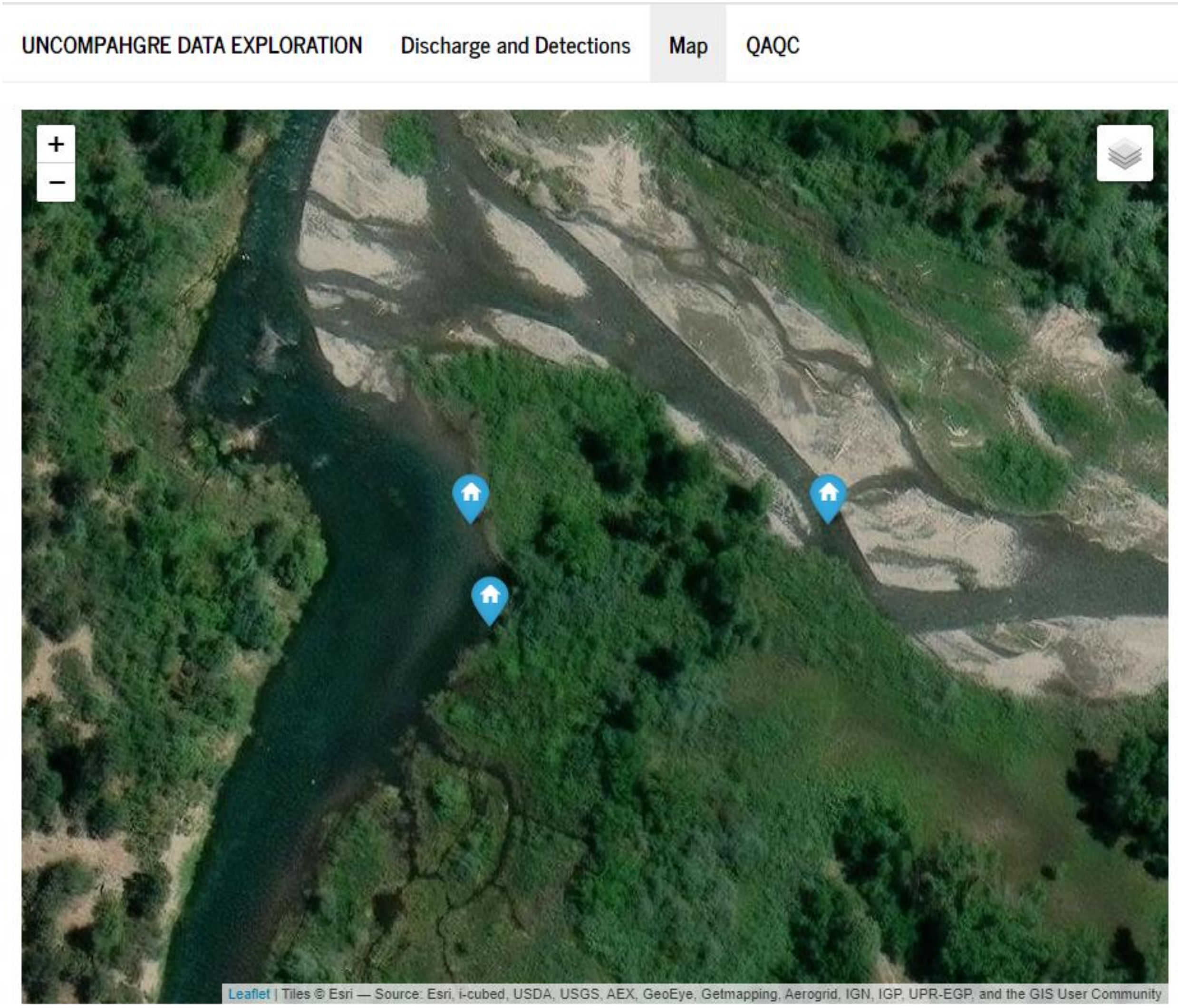
- The data graphed is displayed in the “Count Data (graphed)” tab. Filtered source data is seen in the “All Data” tab.
- Within the tables, you can search for values, display more/less data, and save the data as a .csv or .rds file using the “Save data” button at the bottom of the table.



UNC	989.001030620904	2025-11-11T00:07:53Z
UNC	989.001030620904	2025-11-11T00:08:53Z
UNC	989.001030620904	2025-11-11T00:09:53Z

Map

Interactive leaflet map shows array locations.



QAQC

- There are a few tables to help check for data entry errors. If any of these tables have entries then it's worth examining why that is and correcting them in the tabular data
- These checks are all also currently done in the runscript as well (see "Running RMarkdown Script/Updating the App")

UNCOMPAHGRE DATA EXPLORATION | Discharge and Detections | Map | QAQC

Released Tags with >1 dec_tag Detection | Tags with >1 Entry in Release File | Unknown Tags

Show 10 **entries**

Search:

Tags in Release File that have more than 1 unique tag detection of dec_tag from Biomark data. In other words, most urgent to address tags that are missing digits since they have detections.

Release File tag entry	Number of dec_tag Entries
No data available in table	

Showing 0 to 0 of 0 entries

Previous Next